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CMPU-250 Data and its Discontents

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Project Proposal

Human factors research has been a crucial part of reducing accidents in high pressure and high stakes environments. From aviation to nuclear plants, understanding when things go wrong and why informs researchers on how to find preventative measures. In class, one of our discussions argued that technology and data are not neutral and are filled with biases of the people collecting, analyzing, and interpreting data. In a field that relies heavily on objectivity for human factors analysis, the inability to have neutral data can influence how human factors researchers interpret previous accidents and implement solutions.

One Human Reliability Analysis (HRA) method is a second generation method developed by Erik Hollnagel called the Contextual Control Model (COCOM) of Cognitive Reliability and Error Analysis Method (CREAM). CREAM's goal is to establish "links between the operator's cognitive grade and human error probability," otherwise known as HEPs, and give "the probability of occurrence of the operator's behavior under different control modes" (Pan et al., 2017). COCOM was theorized and built off of CREAM because "human performance was discovered to not be an isolated random action but to be dependent on and controlled by the environment in which the task was completed" (Pan et al., 2017).

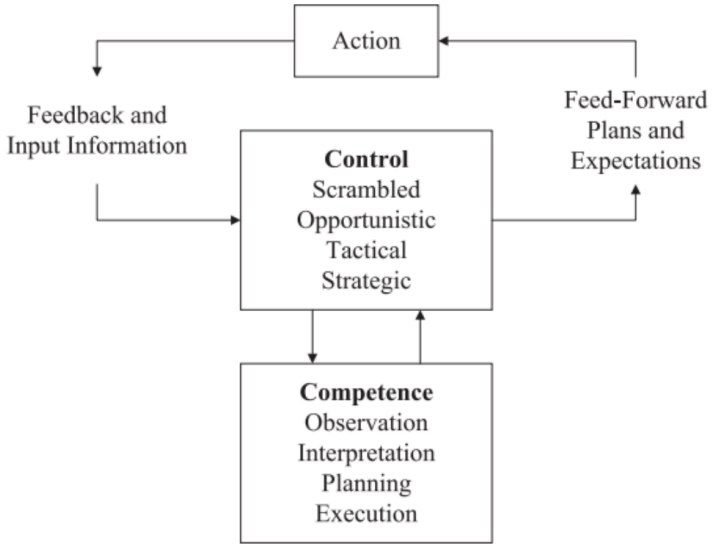


Figure 7. Overview of contextual control model (adapted from Hollnagel[44])

Fig 1. Screenshot of COCOM model (Pan et al., 2017).

The Multi-attribute Technological Accidents Dataset (MATA-D) utilizes COCOM to analyze the interactions between human, technology, and organizational factors and classify the types of error observed in “238 major accidents from different industrial backgrounds” (Moura et al., 2016). Our goal for this project is to understand how biased MATA-D is and in what ways that bias is reflected in the way researchers use and interpret the dataset.

Research Questions:

We have 3 groups of research questions that have qualitative and quantitative components:

- How has the categorization of incidents changed over time? Have there been changes in the number or type of accident? Were the same accidents categorized differently during different years?
- Does this dataset put too much blame on the people instead of the system more than it should? If yes, how are these biases reflected on the AI model trained on this dataset?
- Is the incident adequately described within the categorizations? Are some errors more detrimental than others? Are some errors correlated with others?

Dataset Description, Details, and Rough Sketch:

The MATA-D dataset is a catalogue of 238 major accidents in high-technology industries between the years of 1953 and 2020. In MATA-D, each incident has the mistakes (marked by a 1) categorized under different classifications of error. Each tally has a comment with an explanation of the mistake and why it was categorized where it is. Our first step in analyzing our data, aside from cleaning and formatting these tallied mistakes, will be to appropriately format the mistake comments for qualitative analysis. This allows us to view our dataset as two: one purely quantitative and the other purely qualitative.

Our first research question regarding the change in the categorization and frequency of accidents over time can be answered using this dataset by comparing the reports from different years and analyzing what categories errors were filed under. As such, we can determine whether the total number of incidents or categorizations of a certain type of error have changed over time. Since the data spans such a long time period, it is reasonable to expect that the methods of observing and filing the data may have changed with time, even if the authors categorized all the data in the same year. These changes could impact current interpretations of data, as a higher prevalence of a certain type of error could simply reflect certain methods of classification.

This can also be connected to our second research question regarding whether the dataset puts too much of the blame for accidents on people rather than on institutional or systemic failures. This is something that also may have changed over time. For example, it is possible that an accident caused by a human misusing a piece of technology may have

originally been categorized as a human-caused mistake, whereas a more modern example of a similar mistake may be filed as an institutional problem due to the company providing insufficient training or improper technology.

Categorizational ambiguity such as the example described above could lead to editorializing and the introduction of bias into the dataset. Bad actors or imprudent researchers analyzing this dataset or creating models based on it will then reflect the bias. This observation will be analyzed through the qualitative commented aspect of the dataset, where mistakes of the same kind being categorized differently may allude to our second research question.

Through this qualitative analysis, we can also address the first part of our third research question by analyzing whether categorizations capture the complete picture of an incident. If the categorizations incorrectly describe an incident or leave gaps in the causal chain, this would affect the accuracy and reliability of models trained on the data. Furthermore, we can examine whether the pattern of categorizations more closely matches a domino effect of errors that could be prevented from escalating at any step or a single catastrophic mistake with corollaries. Given that all errors are weighted evenly for a given incident, examining the relative scope and impact of each error can provide further insight into the biases of models trained on the dataset. We supplement this analysis quantitatively by examining the correlation between error categories across incidents. If two or more categories are frequently observed together, it could indicate either of the aforementioned patterns, providing reason to investigate these categorizations further.

Group Contract and Work Agreement

All group members are expected to understand all components of the project, including those they did not directly contribute to. To facilitate this understanding, group members should thoroughly document all work. Time spent on the project should maximize productive progress toward completing the goals of the project as much as possible, rather than focusing effort on perfect work. Group members are expected to meet once a week to work on the project. If a group member is unable to meet during the planned time, they are expected to work independently and communicate progress through text. Work missed due to conflicts, illness, or other unforeseeable circumstances will be made up for later to ensure all members contribute equally and sufficiently to the project.

References

Pan, X., Lin, Y., and He, C. (2017) A review of cognitive models in human reliability analysis. *Quality and Reliability Engineering International, 33*, 1299–1316. doi: 10.1002/qre.2111.

Moura, R., Beer, M., Patelli, E., Lewis, J., and Knoll, F. (2016) *Learning from major accidents to improve system design*. *Safety Science, 84*, 37–45. doi: 10.1016/j.ssci.2015.11.022.